

ZERO-TO-JOB-READY DATA ANALYST 2026

M06 - POWER BI DAX & REPORTING

Measures -> context -> KPIs -> time -> visuals -> interactivity -> QA -> handoff

MODEL TRUST -> MEASURE -> CONTEXT -> VISUAL -> VALIDATE -> DEFEND

DAX never repairs a broken semantic model.

This book is your teacher

- M05 model trust is the prerequisite. If a relationship/filter path is broken, repair the model before writing more DAX.
- Every new DAX word is introduced before use. The formula is always tied to business meaning, active context and an independent validation.
- Videos are visual demonstrations; official Microsoft pages control current function/service/security behavior. The book remains the complete fallback.
- Protected review opens only after a genuine PBIX attempt. After diagnosis, close it and rebuild on changed data/context.
- Publishing, scheduled refresh and service RLS depend on workspace/license/tenant permissions. The course never invents access you do not have.

Normal M06 loop

1. State KPI/business question
2. Name grain + active context
3. Watch/read exact source if useful
4. Follow teacher measure/report once
5. Do guided PBIX task
6. Close support + changed retry
7. Reconcile independently
8. Explain + save evidence

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Module prerequisite and tool map

0. Prerequisite

M05 Power BI Modeling competency: trusted star model, intended 1:* relationships, valid Date dimension, filter propagation, model validation and refresh hygiene. M06 introduces DAX/reporting from there; no advanced DAX is assumed.

Files / tools

- C1_M06_POWERBI_DAX_REPORTING_PRACTICE_DATA_RC1.xlsx + your own PBIX.
- C1_M06_DAX_MEASURE_PATTERNS_RC1.txt - teaching reference, not Gate answers.
- C1_M06_POWERBI_DAX_REPORTING_PRACTICE_GUIDE_RC1.pdf - exact guided/changed route.
- C1_M06_POWERBI_DAX_REPORTING_MASTERY_TRACKER_RC1.xlsx - module mastery control.
- M-PBIR project brief - integrated report/QA/handoff before Gate A.
- Gate A/B/C source workbooks + guides + protected reviews.

Runtime boundary

- DAX/visual/interaction behavior must execute in Power BI Desktop.
- Publish/scheduled refresh/service sharing may be conceptual if access is unavailable.
- Static RLS can be defined/tested with View As in Desktop; service membership needs service access.
- No tenant/license capability is assumed from the workbook alone.

Exact practice, project and Gate route

- PBD1 measures -> PBD2 context -> PBD3 CALCULATE -> PBD4 targets -> PBD5 time intelligence -> PBD6 visual hierarchy -> PBD7 interactivity -> PBD8 report QA -> PBD9 publish/refresh -> PBD10 RLS/handoff.
- Every lesson: learner book -> exact source if useful -> guided PBIX task -> changed task -> independent validation -> explain-back -> append evidence.
- Complete M-PBIR after PBD1-PBD10: executive report + QA page + interaction proof + handoff/security/refresh notes.
- Attempt Gate A independently. A clean pass ends M06.
- Gate B/C are fresh remediation full retests only after prior failure + review + repair + explicit assignment.

Critical-failure examples

Broken model hidden with DAX; wrong denominator context; monthly target used at unsupported grain;
invalid time comparison; report totals contradict QA anchors; interaction changes unintended context;
security/share claims without proof can block a pass.

PBD1 - Measures first: the calculation layer

Practice: matching PBD block in C1_M06_POWERBI_DAX_REPORTING_PRACTICE_GUIDE_RC1.pdf

1. Why this lesson matters

A report can show a believable number while its DAX context, target grain, interaction or validation is wrong. This lesson teaches one controlled reporting competency and how to prove it.

2. What you are learning

Create explicit base measures for count, distinct count, revenue, cost, profit, margin and average value; explain measures vs calculated columns.

3. What you already need / prerequisite

M05 Modeling pass. No DAX measure skill is assumed.

4. Picture it in your head

Fact rows -> explicit measures -> visuals. Reuse [Net Sales], [Profit] and other base measures instead of recreating logic in every visual.

5. New words before we use them

Measure = named DAX calculation evaluated in context. Calculated column = stored row-by-row result. SUMX = iterator that evaluates an expression per row then adds. DISTINCTCOUNT = unique-value count. DIVIDE = safe ratio function.

6. Watch / visual source

RECOMMENDED_VISUAL. Chandoo - Learn 80% of DAX in an Hour | 3:00-15:08. Why use it: See explicit measures, why measures beat auto calculations, and measure-vs-column before typing the base measure set. Internal fallback: PBD1 learner book + DAX reference fully teach the base measures.

[Open primary/reference](#)

PBD1 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Measures are model calculations evaluated when a visual/query asks for them. Calculated columns are row-by-row stored values. Analyst KPIs should normally be explicit measures because business logic stays named, reusable and testable. Use SUM for an existing additive column, SUMX when an expression must be evaluated row by row, DISTINCTCOUNT for business entities, COUNTROWS for rows, and DIVIDE for safe ratios. Teacher action: Create explicit measures: Line Count, Order Count, Customer Count, Total Quantity, Gross Sales, Net Sales, Total Cost, Profit, Margin %, Average Order Value. Save a card/table page and explain SUM vs SUMX and DIVIDE.

8. What you should see / be able to say

Base measures reconcile to protected practice anchors; implicit raw-column aggregation is not used as the final KPI logic.

9. Why it worked

The model is already trusted, the measure/report action answers a stated question under known context, and a second calculation/filter/state check can verify the result. Visual polish is never accepted as proof of numerical correctness.

10. Common beginner mistakes

Using implicit raw-column aggregations as final KPI logic; creating calculated columns for measures that must respond to filters; using SUM when the business calculation is a row expression; raw division with zero denominator.

11. If you get stuck - recovery ladder

If totals look wrong, check the source column, aggregation choice and whether the KPI needs SUM or SUMX. If a ratio divides by zero, use DIVIDE rather than hiding the issue. 1) Restate the business question and active grain/context. 2) Identify the exact measure/visual/filter/relationship involved. 3) Reveal only a structural hint. 4) Open the protected review only after saved evidence of an attempt. 5) Close review and solve a changed retry.

PBD1 - Do alone, validate and master

12. Now you do a small version alone

On a fresh scenario, rebuild the base measure set with different column names and prove one measure using an independent manual/filtered check.

13. Validate your result

Base measures reconcile to protected practice anchors; implicit raw-column aggregation is not used as the final KPI logic.

14. Teach it back in your own words

Without the lesson open, explain measures first: the calculation layer: what problem it solves, the relevant grain/filter context, one validation check and one failure/recovery path.

15. Protected review + changed fresh retry

Save the PBD1 PBIX/evidence first. If repair is needed, open the protected PBD1 review, compare only the weak competency, close it, and complete the changed task in a fresh context. Append the retry; never overwrite the first attempt.

16. Evidence / mastery + exact next route

Save PBIX/version name, relevant measure list/DAX, report/QA screenshots, validation evidence, one failure/recovery note, confidence 1-5 and a short own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> return only to the named weak competency; do not restart the module.

PBD2 - Evaluation context: why the same measure changes

Practice: matching PBD block in C1_M06_POWERBI_DAX_REPORTING_PRACTICE_GUIDE_RC1.pdf

1. Why this lesson matters

A report can show a believable number while its DAX context, target grain, interaction or validation is wrong. This lesson teaches one controlled reporting competency and how to prove it.

2. What you are learning

Predict and test how slicers, visual rows and relationships change the same measure without changing its formula.

3. What you already need / prerequisite

Prerequisite: PBD1 plus the trusted M05 model. The new concept is introduced here before independent use.

4. Picture it in your head

Same measure + different active filters = different valid result. Trace slicer/visual/relationship filters to the fact table.

5. New words before we use them

Evaluation context = conditions under which DAX is evaluated. Filter context = active filters from visual rows/columns, slicers, page/report filters and relationships. Propagation = a relationship carrying filter context to another table.

6. Watch / visual source

RECOMMENDED_REFERENCE. Microsoft Learn - DAX overview | Measures and evaluation-context sections. Why use it: Current official explanation that a measure result depends on the context supplied by the report. Internal fallback: PBD2 learner book + practice guide fully teach filter-context prediction/testing. Optional visual companion: Chandoo - Learn 80% of DAX in an Hour | 19:24-25:12.

PBD2 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Filter context is the set of filters active when a measure is evaluated: visual rows/columns, slicers, page/report filters, relationship propagation and DAX filters. The measure formula can stay identical while the result changes because the question/context changed. When a number surprises you, ask: what filters are active and which relationships carry them to the fact table? Teacher action: Create Category x Region matrix with Net Sales and Profit plus Region/Category slicers. Predict three selections before clicking, test them, and write the observed contexts.

8. What you should see / be able to say

Matrix/slicer results match predictions or corrections; learner can name the active filters that explain each change.

9. Why it worked

The model is already trusted, the measure/report action answers a stated question under known context, and a second calculation/filter/state check can verify the result. Visual polish is never accepted as proof of numerical correctness.

10. Common beginner mistakes

Thinking a measure has one fixed answer; forgetting page/report filters; using a slicer from a disconnected/wrong dimension; blaming DAX before tracing relationship filter propagation.

11. If you get stuck - recovery ladder

If a number changes unexpectedly, remove slicers one at a time, inspect visual/page/report filters, then trace relationship filter flow to the fact table. 1) Restate the business question and active grain/context. 2) Identify the exact measure/visual/filter/relationship involved. 3) Reveal only a structural hint. 4) Open the protected review only after saved evidence of an attempt. 5) Close review and solve a changed retry.

PBD2 - Do alone, validate and master

12. Now you do a small version alone

Given a changed matrix/slicer design, predict five context outcomes before interaction and explain any wrong prediction after testing.

13. Validate your result

Matrix/slicer results match predictions or corrections; learner can name the active filters that explain each change.

14. Teach it back in your own words

Without the lesson open, explain evaluation context: why the same measure changes: what problem it solves, the relevant grain/filter context, one validation check and one failure/recovery path.

15. Protected review + changed fresh retry

Save the PBD2 PBIX/evidence first. If repair is needed, open the protected PBD2 review, compare only the weak competency, close it, and complete the changed task in a fresh context. Append the retry; never overwrite the first attempt.

16. Evidence / mastery + exact next route

Save PBIX/version name, relevant measure list/DAX, report/QA screenshots, validation evidence, one failure/recovery note, confidence 1-5 and a short own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> return only to the named weak competency; do not restart the module.

PBD3 - CALCULATE: deliberately changing context

Practice: matching PBD block in C1_M06_POWERBI_DAX_REPORTING_PRACTICE_GUIDE_RC1.pdf

1. Why this lesson matters

A report can show a believable number while its DAX context, target grain, interaction or validation is wrong. This lesson teaches one controlled reporting competency and how to prove it.

2. What you are learning

Use CALCULATE and REMOVEFILTERS to answer filtered-KPI and share-of-total questions without corrupting the denominator.

3. What you already need / prerequisite

Prerequisite: PBD2 plus the trusted M05 model. The new concept is introduced here before independent use.

4. Picture it in your head

Base measure -> CALCULATE -> modified filter context. Numerator may stay filtered while denominator intentionally removes one breakdown filter.

5. New words before we use them

CALCULATE = evaluate an expression under modified filter context. REMOVEFILTERS = remove specified filters. Numerator = top of ratio. Denominator = bottom of ratio. Share of total = filtered numerator divided by deliberately broader denominator.

6. Watch / visual source

RECOMMENDED_REFERENCE. Microsoft Learn - CALCULATE function | Function semantics and modified filter context. Why use it: Use the current official function semantics while the book teaches denominator/filter reasoning. Internal fallback: PBD3 learner book + DAX reference fully teach CALCULATE/REMOVEFILTERS. Optional visual companion: Chandoo - Learn 80% of DAX in an Hour | 30:56-46:35.

[Open primary/reference](#)

[Open visual companion](#)

PBD3 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

CALCULATE evaluates a measure/expression under modified filter context. Start from trusted base measures and change only the context needed for the question. REMOVEFILTERS is useful when a denominator must ignore a breakdown filter. A common bug is leaving the denominator filtered so every row appears to be 100%. Teacher action: Create East Sales, Electronics Sales, All Category Sales and Category Share %. Validate category shares under a Region slicer and explain the denominator context.

8. What you should see / be able to say

Share denominator behavior is correct; shares reconcile sensibly and do not all show 100%.

9. Why it worked

The model is already trusted, the measure/report action answers a stated question under known context, and a second calculation/filter/state check can verify the result. Visual polish is never accepted as proof of numerical correctness.

10. Common beginner mistakes

Removing too many filters in the denominator; each category share showing 100%; hard-coding a filtered result instead of changing context; using CALCULATE to hide a relationship/model problem.

11. If you get stuck - recovery ladder

If each share row shows 100%, inspect the denominator context and REMOVEFILTERS target. If CALCULATE produces blank, check whether the filter value exists and whether a relationship can propagate it. 1) Restate the business question and active grain/context. 2) Identify the exact measure/visual/filter/relationship involved. 3) Reveal only a structural hint. 4) Open the protected review only after saved evidence of an attempt. 5) Close review and solve a changed retry.

PBD3 - Do alone, validate and master

12. Now you do a small version alone

Build a share-of-total measure for a different dimension and prove the denominator ignores only the intended breakdown filter while respecting another slicer.

13. Validate your result

Share denominator behavior is correct; shares reconcile sensibly and do not all show 100%.

14. Teach it back in your own words

Without the lesson open, explain calculate: deliberately changing context: what problem it solves, the relevant grain/filter context, one validation check and one failure/recovery path.

15. Protected review + changed fresh retry

Save the PBD3 PBIX/evidence first. If repair is needed, open the protected PBD3 review, compare only the weak competency, close it, and complete the changed task in a fresh context. Append the retry; never overwrite the first attempt.

16. Evidence / mastery + exact next route

Save PBIX/version name, relevant measure list/DAX, report/QA screenshots, validation evidence, one failure/recovery note, confidence 1-5 and a short own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> return only to the named weak competency; do not restart the module.

PBD4 - KPI measures and targets

Practice: matching PBD block in C1_M06_POWERBI_DAX_REPORTING_PRACTICE_GUIDE_RC1.pdf

1. Why this lesson matters

A report can show a believable number while its DAX context, target grain, interaction or validation is wrong. This lesson teaches one controlled reporting competency and how to prove it.

2. What you are learning

Build target/variance/attainment KPIs at the target table's real Region + Month grain and reject unsupported daily target comparisons.

3. What you already need / prerequisite

Prerequisite: PBD3 plus the trusted M05 model. The new concept is introduced here before independent use.

4. Picture it in your head

Actual fact (daily/line grain) + target fact-like table (Region + Month grain) -> compatible monthly/region comparison.

5. New words before we use them

KPI = named business performance indicator. Target = planned benchmark. Variance = actual minus target. Attainment = actual divided by target. Target grain = level at which the target is genuinely defined.

6. Watch / visual source

BOOK_LED. Internal lesson - target grain and KPI contract. Why use it: The course-specific Region/Month target grain is better taught against the exact supplied target table than through a generic video. Internal fallback: PBD4 learner book + practice guide are the canonical route.

PBD4 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Targets are facts at a defined grain. If a target table is Region + Month, the model must compare actual vs target at a compatible Region + Month grain unless the business supplies an allocation rule. $\text{Variance} = \text{actual} - \text{target}$; $\text{attainment} = \text{actual} / \text{target}$. A polished card at the wrong grain is still wrong. Teacher action: Relate Regions and Date to Monthly_Targets; create Sales Target, Variance to Target and Target Attainment %. Compare by Region + Month and explain why daily target is unsupported.

8. What you should see / be able to say

Target totals are evaluated at Region + Month grain; actual/target/variance/attainment reconcile to the target table and no unsupported daily target is presented.

9. Why it worked

The model is already trusted, the measure/report action answers a stated question under known context, and a second calculation/filter/state check can verify the result. Visual polish is never accepted as proof of numerical correctness.

10. Common beginner mistakes

Comparing a monthly target to daily actuals as if the target were daily; summing repeated target values at the wrong grain; hiding target blanks as zero; not stating target definition.

11. If you get stuck - recovery ladder

If target repeats or multiplies, stop and restate target grain. Check target-table key uniqueness and the Date/Region relationships before changing DAX. 1) Restate the business question and active grain/context. 2) Identify the exact measure/visual/filter/relationship involved. 3) Reveal only a structural hint. 4) Open the protected review only after saved evidence of an attempt. 5) Close review and solve a changed retry.

PBD4 - Do alone, validate and master

12. Now you do a small version alone

Use a different target table at Branch + Month grain. Build actual/target/variance/attainment and reject one intentionally wrong daily visualization.

13. Validate your result

Target totals are evaluated at Region + Month grain; actual/target/variance/attainment reconcile to the target table and no unsupported daily target is presented.

14. Teach it back in your own words

Without the lesson open, explain kpi measures and targets: what problem it solves, the relevant grain/filter context, one validation check and one failure/recovery path.

15. Protected review + changed fresh retry

Save the PBD4 PBIX/evidence first. If repair is needed, open the protected PBD4 review, compare only the weak competency, close it, and complete the changed task in a fresh context. Append the retry; never overwrite the first attempt.

16. Evidence / mastery + exact next route

Save PBIX/version name, relevant measure list/DAX, report/QA screenshots, validation evidence, one failure/recovery note, confidence 1-5 and a short own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> return only to the named weak competency; do not restart the module.

PBD5 - Time intelligence that you can defend

Practice: matching PBD block in C1_M06_POWERBI_DAX_REPORTING_PRACTICE_GUIDE_RC1.pdf

1. Why this lesson matters

A report can show a believable number while its DAX context, target grain, interaction or validation is wrong. This lesson teaches one controlled reporting competency and how to prove it.

2. What you are learning

Create YTD, prior-year and previous-month measures using a valid Date table and defend boundary periods such as January.

3. What you already need / prerequisite

Prerequisite: PBD4 plus the trusted M05 model. The new concept is introduced here before independent use.

4. Picture it in your head

Date table -> active fact-date relationship -> DATESYTD / SAMEPERIODLASTYEAR / DATEADD -> validated boundary comparisons.

5. New words before we use them

Time intelligence = calculations across calendar periods. YTD = year-to-date. Prior year = comparable earlier-year period. Previous month = immediately prior month. Boundary period = period such as January where the comparison crosses a year boundary.

6. Watch / visual source

RECOMMENDED_VISUAL. Chandoo - FREE Adv. Masterclass on Power BI - Create a Sales Analytics Dashboard | 17:43-27:22. Why use it: See calendar/time-intelligence measures in a real dashboard before building YTD/previous-month/prior-year checks on the course Date table. Internal fallback: PBD5 learner book + DAX reference teach exact formulas and boundary checks.

PBD5 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Time intelligence depends on a proper Date table: one row per date, coverage across the required analysis period and an active relationship to the fact date. YTD, prior-year and prior-month measures are context transformations, not hard-coded lookups. January is a key validation period because previous month crosses a year boundary. Teacher action: Create Sales YTD, Sales PY, YoY %, Previous Month Sales and MoM %. Build a monthly 2025-2026 line chart and test January 2026 against December 2025 plus one prior-year period.

8. What you should see / be able to say

January 2026 previous month equals December 2025; prior-year measure uses the same comparable period; YTD increases logically across months.

9. Why it worked

The model is already trusted, the measure/report action answers a stated question under known context, and a second calculation/filter/state check can verify the result. Visual polish is never accepted as proof of numerical correctness.

10. Common beginner mistakes

Using Month Name without chronological key; invalid/missing Date relationship; comparing incomplete periods without saying so; January previous-month logic failing across year boundary; raw division by blank/zero prior period.

11. If you get stuck - recovery ladder

If previous month/YTD is blank or strange, validate the Date table coverage/continuity, the active Date relationship and the current date context before rewriting the measure. 1) Restate the business question and active grain/context. 2) Identify the exact measure/visual/filter/relationship involved. 3) Reveal only a structural hint. 4) Open the protected review only after saved evidence of an attempt. 5) Close review and solve a changed retry.

PBD5 - Do alone, validate and master

12. Now you do a small version alone

Use a different two-year dataset to build YTD/PY/YoY/previous-month/MoM and prove a year-boundary month plus one missing-prior-period blank is legitimate.

13. Validate your result

January 2026 previous month equals December 2025; prior-year measure uses the same comparable period; YTD increases logically across months.

14. Teach it back in your own words

Without the lesson open, explain time intelligence that you can defend: what problem it solves, the relevant grain/filter context, one validation check and one failure/recovery path.

15. Protected review + changed fresh retry

Save the PBD5 PBIX/evidence first. If repair is needed, open the protected PBD5 review, compare only the weak competency, close it, and complete the changed task in a fresh context. Append the retry; never overwrite the first attempt.

16. Evidence / mastery + exact next route

Save PBIX/version name, relevant measure list/DAX, report/QA screenshots, validation evidence, one failure/recovery note, confidence 1-5 and a short own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> return only to the named weak competency; do not restart the module.

PBD6 - Visual selection and page hierarchy

Practice: matching PBD block in C1_M06_POWERBI_DAX_REPORTING_PRACTICE_GUIDE_RC1.pdf

1. Why this lesson matters

A report can show a believable number while its DAX context, target grain, interaction or validation is wrong. This lesson teaches one controlled reporting competency and how to prove it.

2. What you are learning

Choose visuals that match trend/ranking/detail questions and build a focused executive page with clear reading order and accessible labels.

3. What you already need / prerequisite

Prerequisite: PBD5 plus the trusted M05 model. The new concept is introduced here before independent use.

4. Picture it in your head

Business question -> visual type -> page hierarchy -> labels/formats/accessibility -> interpretation test.

5. New words before we use them

Visual encoding = how values map to position/length/color/etc. Hierarchy = reading order and emphasis. KPI card = compact headline value. Trend = ordered time movement. Ranking = ordered category comparison. Accessibility = readable labels/contrast/meaning beyond color alone.

6. Watch / visual source

RECOMMENDED_VISUAL. Chandoo - FREE Adv. Masterclass on Power BI - Create a Sales Analytics Dashboard | 27:22-44:26. Why use it: See wireframe, page hierarchy, KPI cards and formatting choices before building the executive page. Internal fallback: PBD6 learner book + practice guide teach question-to-visual selection and accessibility checks.

PBD6 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Visual choice is part of analysis. Cards communicate a few headline KPIs; line charts show ordered time; bars compare/rank categories; matrices support detail; scatterplots show numeric relationships. Page hierarchy should make the main question obvious before supporting detail. Avoid decoration that carries no information and do not rely on color alone. Teacher action: Build a 16:9 Executive Summary: 4-5 KPI cards, monthly trend, Category bar, Region bar and one supporting matrix/table. Save a three-sentence design rationale.

8. What you should see / be able to say

Each visual answers a named question, formats are consistent, reading order is obvious and the page does not rely on color alone.

9. Why it worked

The model is already trusted, the measure/report action answers a stated question under known context, and a second calculation/filter/state check can verify the result. Visual polish is never accepted as proof of numerical correctness.

10. Common beginner mistakes

Too many visuals/KPIs; chart chosen by decoration rather than question; unreadable tiny labels; color is the only meaning; different pages use inconsistent definitions.

11. If you get stuck - recovery ladder

If the page feels crowded, remove visuals that duplicate another question, reduce slicers and keep 3-5 headline KPIs. If interpretation depends on color alone, add labels/signs. 1) Restate the business question and active grain/context. 2) Identify the exact measure/visual/filter/relationship involved. 3) Reveal only a structural hint. 4) Open the protected review only after saved evidence of an attempt. 5) Close review and solve a changed retry.

PBD6 - Do alone, validate and master

12. Now you do a small version alone

Redesign a cluttered executive page for a different stakeholder: keep only the questions that support one decision and defend every visual kept/removed.

13. Validate your result

Each visual answers a named question, formats are consistent, reading order is obvious and the page does not rely on color alone.

14. Teach it back in your own words

Without the lesson open, explain visual selection and page hierarchy: what problem it solves, the relevant grain/filter context, one validation check and one failure/recovery path.

15. Protected review + changed fresh retry

Save the PBD6 PBIX/evidence first. If repair is needed, open the protected PBD6 review, compare only the weak competency, close it, and complete the changed task in a fresh context. Append the retry; never overwrite the first attempt.

16. Evidence / mastery + exact next route

Save PBIX/version name, relevant measure list/DAX, report/QA screenshots, validation evidence, one failure/recovery note, confidence 1-5 and a short own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> return only to the named weak competency; do not restart the module.

PBD7 - Interactivity without confusion

Practice: matching PBD block in C1_M06_POWERBI_DAX_REPORTING_PRACTICE_GUIDE_RC1.pdf

1. Why this lesson matters

A report can show a believable number while its DAX context, target grain, interaction or validation is wrong. This lesson teaches one controlled reporting competency and how to prove it.

2. What you are learning

Configure useful slicers, edit interactions, drillthrough and report-page tooltips; prove interactivity does not create contradictory context.

3. What you already need / prerequisite

Prerequisite: PBD6 plus the trusted M05 model. The new concept is introduced here before independent use.

4. Picture it in your head

User action -> filter/highlight/drillthrough/tooltip -> changed context -> measure recalculation -> validation of intended behavior.

5. New words before we use them

Interaction = one visual/slicer affecting another. Drillthrough = open a detail page for selected context. Tooltip page = compact supporting context on hover. Edit interactions = choose filter/highlight/none behavior among visuals.

6. Watch / visual source

RECOMMENDED_VISUAL. Chandoo - FREE Adv. Masterclass on Power BI - Create a Sales Analytics Dashboard | 1:25:35-1:44:16. Why use it: See bookmarks/filter panel/tooltips/interactivity in motion before testing interactions deliberately. Internal fallback: PBD7 learner book + practice guide teach slicer/edit-interaction/drillthrough/tooltip validation.

PBD7 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Interactivity must reduce effort, not hide context. Use only high-value slicers. Explicitly inspect Edit interactions. Drillthrough is for a selected entity/detail path; report-page tooltips give compact supporting context. A visual interaction is correct only when its behavior matches the business question and validation totals remain consistent. Teacher action: Add only useful Date/Region/Category slicers, inspect Edit interactions, create one drillthrough page and one report-page tooltip, then save one verified interaction state.

8. What you should see / be able to say

Filter/highlight/none behavior matches intent; drillthrough receives the selected context; tooltip values do not contradict the source visual.

9. Why it worked

The model is already trusted, the measure/report action answers a stated question under known context, and a second calculation/filter/state check can verify the result. Visual polish is never accepted as proof of numerical correctness.

10. Common beginner mistakes

Every slicer affects every visual by accident; drillthrough field missing; tooltip contradicts source context; bookmark captures unwanted filter/data state; interactivity added without a business purpose.

11. If you get stuck - recovery ladder

If a slicer affects the wrong visual, use Edit interactions and test filter/highlight/none explicitly. If drillthrough is unavailable, verify the drillthrough field and selected context. 1) Restate the business question and active grain/context. 2) Identify the exact measure/visual/filter/relationship involved. 3) Reveal only a structural hint. 4) Open the protected review only after saved evidence of an attempt. 5) Close review and solve a changed retry.

PBD7 - Do alone, validate and master

12. Now you do a small version alone

Configure a changed report where one slicer should not affect one diagnostic visual; prove Edit interactions, drillthrough context and tooltip values are correct.

13. Validate your result

Filter/highlight/none behavior matches intent; drillthrough receives the selected context; tooltip values do not contradict the source visual.

14. Teach it back in your own words

Without the lesson open, explain interactivity without confusion: what problem it solves, the relevant grain/filter context, one validation check and one failure/recovery path.

15. Protected review + changed fresh retry

Save the PBD7 PBIX/evidence first. If repair is needed, open the protected PBD7 review, compare only the weak competency, close it, and complete the changed task in a fresh context. Append the retry; never overwrite the first attempt.

16. Evidence / mastery + exact next route

Save PBIX/version name, relevant measure list/DAX, report/QA screenshots, validation evidence, one failure/recovery note, confidence 1-5 and a short own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> return only to the named weak competency; do not restart the module.

PBD8 - Report validation: prove the story is true

Practice: matching PBD block in C1_M06_POWERBI_DAX_REPORTING_PRACTICE_GUIDE_RC1.pdf

1. Why this lesson matters

A report can show a believable number while its DAX context, target grain, interaction or validation is wrong. This lesson teaches one controlled reporting competency and how to prove it.

2. What you are learning

Create a QA/validation page and independently reconcile totals, dimensional breakdowns, targets, time measures and interaction behavior.

3. What you already need / prerequisite

Prerequisite: PBD7 plus the trusted M05 model. The new concept is introduced here before independent use.

4. Picture it in your head

Source anchors -> model measures -> breakdown totals -> time/target checks -> interaction checks -> evidence log.

5. New words before we use them

QA page = hidden/visible validation area containing trusted controls. Reconciliation = independent agreement between report output and a second calculation/path. Anchor = expected control value. Interaction-state test = validate before/after a slicer or drill action.

6. Watch / visual source

BOOK_LED. Internal lesson - report QA and reconciliation. Why use it: Validation must use the exact protected anchors and course model; a generic video would add less value than the controlled QA page. Internal fallback: PBD8 learner book + practice guide are the canonical route.

PBD8 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Validation is a deliverable. Reconcile source rows, base measures, dimensional totals, target/time calculations, interactions and cross-page consistency. Test both grand total and breakdown grain because context bugs can hide at one level. Keep a QA page/log with Expected, Actual, Status and the evidence used. Teacher action: Create a QA page/log and reconcile line count, order count, customer count, Net Sales and Profit. Add one target check, one time-intelligence check and one slicer-interaction check.

8. What you should see / be able to say

At least seven validation checks have Expected/Actual/Status and include base, breakdown, target, time and interaction evidence.

9. Why it worked

The model is already trusted, the measure/report action answers a stated question under known context, and a second calculation/filter/state check can verify the result. Visual polish is never accepted as proof of numerical correctness.

10. Common beginner mistakes

Checking only grand total; QA page uses the same wrong measure and therefore is not independent; ignoring target/time mismatches; interaction tests not documented; validating after presentation claims are already written.

11. If you get stuck - recovery ladder

If a grand total matches but categories do not, validate at the breakdown grain. If Expected vs Actual differs, isolate one measure/context at a time rather than rebuilding the report. 1) Restate the business question and active grain/context. 2) Identify the exact measure/visual/filter/relationship involved. 3) Reveal only a structural hint. 4) Open the protected review only after saved evidence of an attempt. 5) Close review and solve a changed retry.

PBD8 - Do alone, validate and master

12. Now you do a small version alone

Receive a report with one planted wrong KPI/context behavior. Use the validation ladder to identify the failing layer and document Expected -> Actual -> Cause -> Fix.

13. Validate your result

At least seven validation checks have Expected/Actual/Status and include base, breakdown, target, time and interaction evidence.

14. Teach it back in your own words

Without the lesson open, explain report validation: prove the story is true: what problem it solves, the relevant grain/filter context, one validation check and one failure/recovery path.

15. Protected review + changed fresh retry

Save the PBD8 PBIX/evidence first. If repair is needed, open the protected PBD8 review, compare only the weak competency, close it, and complete the changed task in a fresh context. Append the retry; never overwrite the first attempt.

16. Evidence / mastery + exact next route

Save PBIX/version name, relevant measure list/DAX, report/QA screenshots, validation evidence, one failure/recovery note, confidence 1-5 and a short own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> return only to the named weak competency; do not restart the module.

PBD9 - Publish, workspaces, refresh and sharing basics

Practice: matching PBD block in C1_M06_POWERBI_DAX_REPORTING_PRACTICE_GUIDE_RC1.pdf

1. Why this lesson matters

A report can show a believable number while its DAX context, target grain, interaction or validation is wrong. This lesson teaches one controlled reporting competency and how to prove it.

2. What you are learning

Explain publish/workspace/refresh/sharing workflow accurately without inventing tenant or license capabilities; demonstrate Desktop refresh when service publishing is unavailable.

3. What you already need / prerequisite

Prerequisite: PBD8 plus the trusted M05 model. The new concept is introduced here before independent use.

4. Picture it in your head

Desktop PBIX -> Publish (when allowed) -> workspace report + semantic model -> credentials/gateway/source rules -> refresh -> sharing within permissions.

5. New words before we use them

Publish = send Desktop semantic model/report to Power BI service workspace. Workspace = service collaboration container. Semantic model = published model used by reports. Refresh = update from source. Gateway = bridge often required for on-premises sources. License/tenant = account capabilities that vary.

6. Watch / visual source

RECOMMENDED_REFERENCE. Microsoft Learn - Publish semantic models and reports from Power BI Desktop | Publish workflow, republish and limitations. Why use it: Current official workflow; avoids teaching stale tenant/workspace UI as universal. Internal fallback: PBD9 learner book explains Desktop refresh, publish concepts and capability/tenant caveats.

PBD9 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Publishing sends the Desktop report/semantic model to a Power BI workspace, subject to account and tenant permissions. Browser visual refresh is not the same as imported-data refresh. Scheduled/data refresh updates the semantic model from sources and may require credentials/gateway depending on source location. The course never requires buying a license. Teacher action: If allowed, publish to My workspace and inspect report/semantic model; otherwise demonstrate Desktop Refresh and write the publish/refresh workflow accurately.

8. What you should see / be able to say

Workflow distinguishes publishing from data refresh and states account/tenant limitations accurately; no fabricated service evidence.

9. Why it worked

The model is already trusted, the measure/report action answers a stated question under known context, and a second calculation/filter/state check can verify the result. Visual polish is never accepted as proof of numerical correctness.

10. Common beginner mistakes

Claiming scheduled refresh/workspace sharing works without access; confusing Desktop Refresh with service scheduled refresh; promising license/tenant capabilities; forgetting gateway/credentials dependencies.

11. If you get stuck - recovery ladder

If publish/refresh is unavailable, do not invent service success. Demonstrate Desktop refresh and write the correct service workflow/licensing caveat. 1) Restate the business question and active grain/context. 2) Identify the exact measure/visual/filter/relationship involved. 3) Reveal only a structural hint. 4) Open the protected review only after saved evidence of an attempt. 5) Close review and solve a changed retry.

PBD9 - Do alone, validate and master

12. Now you do a small version alone

Write a deployment/refresh handoff for a new source location, explicitly separating Desktop refresh, semantic-model refresh, credentials, gateway and sharing permissions.

13. Validate your result

Workflow distinguishes publishing from data refresh and states account/tenant limitations accurately; no fabricated service evidence.

14. Teach it back in your own words

Without the lesson open, explain publish, workspaces, refresh and sharing basics: what problem it solves, the relevant grain/filter context, one validation check and one failure/recovery path.

15. Protected review + changed fresh retry

Save the PBD9 PBIX/evidence first. If repair is needed, open the protected PBD9 review, compare only the weak competency, close it, and complete the changed task in a fresh context. Append the retry; never overwrite the first attempt.

16. Evidence / mastery + exact next route

Save PBIX/version name, relevant measure list/DAX, report/QA screenshots, validation evidence, one failure/recovery note, confidence 1-5 and a short own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> return only to the named weak competency; do not restart the module.

PBD10 - RLS basics, handoff and defense

Practice: matching PBD block in C1_M06_POWERBI_DAX_REPORTING_PRACTICE_GUIDE_RC1.pdf

1. Why this lesson matters

A report can show a believable number while its DAX context, target grain, interaction or validation is wrong. This lesson teaches one controlled reporting competency and how to prove it.

2. What you are learning

Create/test a simple static RLS role, distinguish security from slicers/page hiding, and produce a professional handoff/defense package.

3. What you already need / prerequisite

Prerequisite: PBD9 plus the trusted M05 model. The new concept is introduced here before independent use.

4. Picture it in your head

Role filter -> model rows -> View as roles test -> handoff notes/limitations -> defense of KPI, context, validation, refresh and security decisions.

5. New words before we use them

RLS = row-level security that filters rows for a role/user. Role = named security rule set. View As = Desktop role test. Role mapping = assign users/groups in service. Handoff = package of PBIX, definitions, validation and operating notes for another analyst.

6. Watch / visual source

RECOMMENDED_VISUAL. Guy in a Cube - How to Setup Row-Level Security (RLS) in Power BI | 0:00-4:30. Why use it: Short visual walkthrough of role creation, View As and service role mapping before the course handoff/security exercise. Internal fallback: PBD10 learner book + current Microsoft RLS reference fully teach the security distinction and test route.

PBD10 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

RLS restricts rows a user can query through model roles; a slicer or hidden page is not security. For beginner practice, a static role such as `Regions[Region] = "East"` is enough. A professional handoff also includes KPI definitions, PBIX/version, sources/refresh notes, validation results, RLS assumptions, known limitations and portfolio screenshots. Teacher action: Create static East Only role, use View as roles, prove other regions disappear, then write the final handoff checklist and a 90-second project defense.

8. What you should see / be able to say

RLS role visibly restricts rows under View as roles; handoff includes KPI definitions, refresh/source notes, validation and security assumptions.

9. Why it worked

The model is already trusted, the measure/report action answers a stated question under known context, and a second calculation/filter/state check can verify the result. Visual polish is never accepted as proof of numerical correctness.

10. Common beginner mistakes

Treating hidden pages or slicers as security; role filter placed on the wrong table; not using View As; assuming workspace Admin/Member behavior equals Viewer RLS enforcement; sharing PBIX without metric/security/refresh notes.

11. If you get stuck - recovery ladder

If RLS test does not restrict rows, verify the role expression and relationship path, then use View as roles. Do not treat hidden pages or slicers as security. 1) Restate the business question and active grain/context. 2) Identify the exact measure/visual/filter/relationship involved. 3) Reveal only a structural hint. 4) Open the protected review only after saved evidence of an attempt. 5) Close review and solve a changed retry.

PBD10 - Do alone, validate and master

12. Now you do a small version alone

Create a different static RLS role (for example one Club/Zone/Clinic), validate it with View as roles, and explain why the role is security while a slicer is not.

13. Validate your result

RLS role visibly restricts rows under View as roles; handoff includes KPI definitions, refresh/source notes, validation and security assumptions.

14. Teach it back in your own words

Without the lesson open, explain rls basics, handoff and defense: what problem it solves, the relevant grain/filter context, one validation check and one failure/recovery path.

15. Protected review + changed fresh retry

Save the PBD10 PBIX/evidence first. If repair is needed, open the protected PBD10 review, compare only the weak competency, close it, and complete the changed task in a fresh context. Append the retry; never overwrite the first attempt.

16. Evidence / mastery + exact next route

Save PBIX/version name, relevant measure list/DAX, report/QA screenshots, validation evidence, one failure/recovery note, confidence 1-5 and a short own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> return only to the named weak competency; do not restart the module.

M-PBIR - Integrated executive report, QA and handoff

1. Why this project matters

A junior analyst must deliver more than isolated measures. The employer needs a report whose KPI definitions, context, targets, time comparisons, interactions, QA and operating notes can survive handoff.

2. What you are building

A two-page Power BI deliverable on RetailPulse: Executive page + QA page, plus a small drillthrough/tooltip path, measure dictionary, refresh/publish notes and a static RLS role/test where available.

3. Prerequisite

PBD1-PBD10 fresh evidence. M05 model must remain unchanged/trusted unless you document a justified model repair before continuing.

4. Picture it in your head

Trusted model -> explicit base measures -> target/time branches -> executive visuals -> interactions -> QA anchors -> operating/security handoff. If a visual fails validation, trace back through measure -> context -> relationship -> source.

5. New words before we use them

Measure dictionary = measure name/formula/meaning/format/validation. Executive page = concise decision page. QA page = validation controls. Operating note = refresh/source/workspace assumptions. Security note = RLS role/filter/test evidence and limitations.

6. Watch / visual source

No new required video. Reuse only the lesson-specific source for a named weak skill. Project mastery must come from your PBIX, reconciliations and defense.

M-PBIR - Worked micro-example and setup

7. Follow with me once - fully worked teacher demonstration

Start with [Net Sales], [Profit], [Margin %]. Put them on cards. Add Region slicer and Category x Region matrix. Before clicking East, predict the cards. Click East, record the changed values and active context. On QA page create a table by Region using the same measures and compare the East row to the filtered card. This proves one simple measure/context/visual/QA chain before the full project.

8. What you should see / be able to say

The same measure gives different valid values under different filters, but the filtered card and the matching QA table row reconcile. You can name the active filters rather than saying 'Power BI changed the number'.

9. Why it worked

One base measure is reused everywhere, the filter context is visible, and the QA page supplies a second reporting path. That is safer than separate hard-coded calculations for each card.

10. Common beginner mistakes

Building all pages before validating base measures; target at daily grain; hiding a failed KPI behind formatting; using unrelated slicers; no QA page; service/RLS claims copied from a tutorial without actual access.

11. If you get stuck - recovery ladder

Return to the earliest layer: M05 model -> base measure -> context -> target/time branch -> visual -> interaction -> QA. Validate one simple card/table before rebuilding the whole page.

M-PBIR - Full independent build

12. Now you do the project

- Create a measure table/folder arrangement and base measures for count, distinct count, sales, cost, profit, margin and average value.
- Add at least one deliberate CALCULATE KPI and one share-of-total denominator using REMOVEFILTERS.
- Create target/variance/attainment at supported Region + Month grain; show blank/not-applicable where target is unsupported.
- Create previous month, prior year and YTD measures; validate January boundary behavior.
- Executive page: 3-5 KPI cards, one time trend, one ranking/category comparison and one useful detail table/matrix.
- Use only useful slicers. Configure one intentional interaction exception, one drillthrough page and one report-page tooltip.
- QA page: protected practice anchors, region/category breakdown, target check, Jan/Dec/prior-year controls and one interaction-state proof.
- Publish only if your workspace/license permits. Otherwise document the exact conceptual steps and prove Desktop Refresh.
- Create/test a simple static RLS role with View As if Desktop supports it; document that slicers/page hiding are not security.
- Create a handoff note: source, refresh dependency, measures, target grain, known limitations, RLS role, validation checklist and file/version.

13. Validate

Required: base measure anchors; one target reconciliation; Jan 2026 vs Dec 2025 and Jan 2025; top Region and Category; one slicer/edit-interaction state; refresh evidence; and RLS View As evidence or a documented 'not available' limitation. Failed controls block presentation claims.

M-PBIR - Explain, evidence and project rubric

14. Teach it back in your own words

Give a 2-minute defense: model prerequisite -> base measures -> one context modification -> target grain -> time comparison -> visual choices -> interaction behavior -> QA controls -> refresh/publish boundary -> RLS/security boundary -> recommendation/limitation.

15. Protected review + changed fresh retry

Save the full project first. If one competency is weak, open only that lesson/project review. Close answer material and rebuild the affected KPI/page/interaction on changed context before mastery.

16. Evidence / mastery + exact next route

Critical pass: model remains trusted; explicit reusable measures; ratios safe; CALCULATE denominator correct; target grain defended; time boundary validated; visuals match questions; interactions behave intentionally; QA anchors reconcile; publish/refresh claims are evidence-based; RLS is tested or explicitly unavailable; handoff is complete; learner can defend without review. PASS -> Gate A. PARTIAL/FAIL -> targeted repair + changed project evidence before Gate A.

End-of-module mastery checklist

- PBD1: Explicit reusable measures; SUM vs SUMX; distinct count; DIVIDE; measure vs calculated column.
- PBD2: Predict/filter-context changes and trace slicer/visual/relationship filters.
- PBD3: Use CALCULATE/REMOVEFILTERS deliberately and defend numerator/denominator context.
- PBD4: Target/variance/attainment only at supported target grain.
- PBD5: YTD/previous-month/prior-year with valid Date table and January boundary checks.
- PBD6: Choose visuals by question and create clear hierarchy/accessibility.
- PBD7: Test slicers, Edit interactions, drillthrough and report-page tooltips.
- PBD8: Build QA page and independently reconcile totals, breakdowns, targets, time and interactions.
- PBD9: Distinguish Desktop refresh vs service publish/scheduled refresh and state access limitations.
- PBD10: Create/test static RLS basics; distinguish security from presentation; produce handoff evidence.
- M-PBIR: Defend the complete report from model -> DAX -> context -> visuals -> QA -> operations/security.

Gate readiness

Attempt Gate A only after M-PBIR and the protected practice anchors reconcile.

Clean Gate A pass ends M06; B/C are remediation full retests only.

Revision / DAX-report quick reference

Measure	Dynamic calculation evaluated under context.
Calculated column	Stored row-by-row value; not a substitute for context-sensitive KPI measure.
SUMX	Evaluate row expression then sum.
DIVIDE	Safe ratio with denominator handling.
Filter context	Slicers/visual filters/page/report filters/relationships affecting evaluation.
CALCULATE	Evaluate under modified filter context.
REMOVEFILTERS	Broaden only the intended denominator/filter scope.
Target grain	Level where target is defined; do not invent finer-grain target.
Time intelligence	Valid Date table + relationship + boundary validation.
QA page	Independent controls before trusting presentation.
RLS	Row security; not slicer/page hiding. Test with View As.

Next route: M-PBIR -> Gate A -> M07 Python for Data Analysts only after M06 competency pass.